

Non-Parametric Rehearsal Learning via Conditional Mean Embeddings

Anonymous Authors¹

Abstract

In machine learning, a critical class of decision-related problems concerns preventing predicted undesirable outcomes, referred to as the *avoiding undesired future* (AUF) problem. To address this, the *rehearsal learning* framework has been proposed to model influence relations for effective decisions. However, existing rehearsal methods rely on restrictive parametric assumptions such as linear systems or additive noise, limiting their practical applicability. In this paper, we propose the first non-parametric rehearsal learning approach for AUF without assuming specific functional forms of data generation processes. Specifically, we use kernel machinery to reformulate the AUF objective into a unified representation that disentangles desirability modeling from action-induced distributional changes. To handle the discontinuity of desirability indicator, we present a smooth Probit surrogate and provide an approximation error bound. Meanwhile, we capture the action-induced changes via conditional mean embeddings, and develop a kernel ridge regression based nested estimator for AUF objective with consistency guarantees. Such a formulation naturally accommodates nonlinear systems and non-additive noise, and empirical results demonstrate the effectiveness and flexibility of our approach.

1. Introduction

Machine learning (ML) have achieved great success in various real-world prediction tasks (LeCun et al., 2015). Instead of solely focusing on prediction, Zhou (2022) emphasizes another important issue, *i.e.*, if an ML model predicts that something undesired is going to occur, how to find effective actions to prevent it from happening. This problem is known as *avoiding undesired future* (AUF) problem (Zhou, 2022).

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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Consider a motivating example of bank lending illustrated in Fig. 1, where the bank seeks to proactively optimize its loan approval strategy. The objective is to maximize the probability that the target outcomes $\mathbf{Y}$ (*e.g.*, repayment rate and return on investment (ROI)) satisfy specific expectations, formalized as falling into a desired region $\mathcal{S}$ (*e.g.*, $\{\mathbf{y} \mid \text{repayment rate} \geq \alpha, \text{ROI} \geq \beta\}$). In practice, the bank typically observes user profiles $\mathbf{X}$ (*e.g.*, credit score and debt-to-income ratio) to predict $\mathbf{Y}$. If the model anticipates a high likelihood of an undesired future (*i.e.*, $\mathbf{Y} \notin \mathcal{S}$), it becomes imperative to actively alter actionable intermediate variables $\mathbf{Z}_A$ (*e.g.*, interest rate) to steer $\mathbf{Y}$ towards $\mathcal{S}$.

A main challenge in such scenario lies in the fact that historical records are collected observationally, where the values of $\mathbf{Z}_A$ could be confounded by unactionable factors (*e.g.*, market fluctuations) that affect both $\mathbf{Z}_A$ and $\mathbf{Y}$. Consequently, the data distribution under such active alteration differs significantly from the conditional distribution on the historical data. Furthermore, in high-stakes scenarios like financial lending, actively exploring alterations via trial-and-error is prohibitive due to the immense risk of financial loss. Thus, finding optimal decision alterations to maximize the probability of $\mathbf{Y} \in \mathcal{S}$ (*i.e.*, AUF probability) solely based on historical data becomes the core objective of AUF problem.

To address this, one solution is by *rehearsal learning* framework (Zhou, 2022), which leverages fine-grained structural information, including both direct and bi-directed influence, to guide the selection of decision alterations. While existing rehearsal-based methods have made progress, particularly under assumptions of linear additive noise models (Qin et al., 2023; Du et al., 2025), such assumptions rarely hold in complex real-world systems, posing practical challenges. Fundamentally, these methods are parametric (and predominantly assumes linear Gaussian) that heavily relies on the alignment between the true data generation mechanism and the hypothesized parametric forms. Once such alignment is not achievable, the resulting decision strategy suffers from severe performance degradation due to model misspecification. While Qin et al. (2025) attempts to accommodate non-linear scenarios, it remains parametric, and resorts to a heuristic objective which can deviate substantially from the true AUF probability. Consequently, this approach could not guarantee that the suggested decision aligns with the true AUF goal, often yielding suboptimal empirical results.

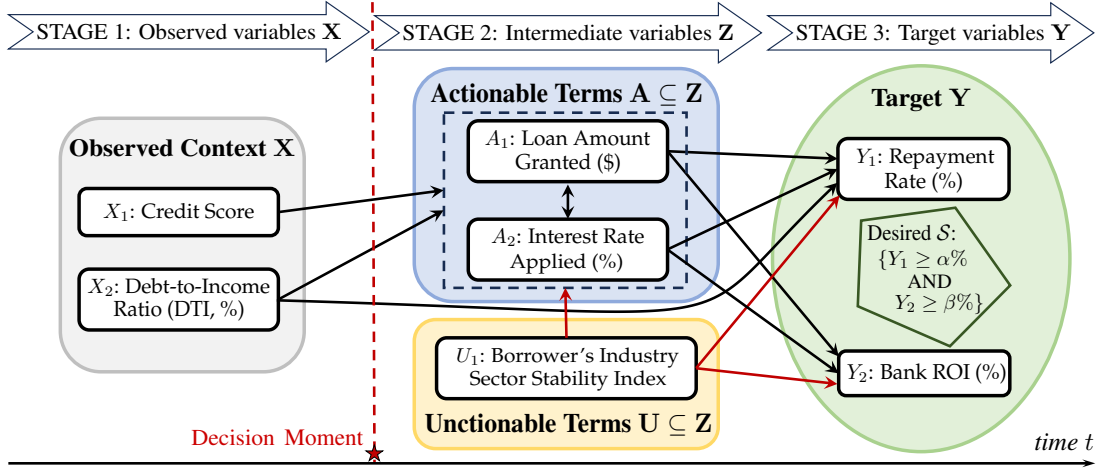


Figure 1. An example illustrating the bank lending AUF scenario.

Considering the limitations, in this paper, we propose the first non-parametric rehearsal learning framework that addresses AUF without relying on specific functional form assumptions. Our key idea is to use kernel machinery to make the AUF objective both *estimable* from observational data and *optimizable* over potential actions. Specifically, we leverage conditional mean embeddings (CMEs) to reformulate the original AUF probability maximization as maximizing an inner product between a desirability function and an action-dependent embedding of target variables Y within a reproducing kernel hilbert space (RKHS). To handle the discontinuity of the desirability indicator $\mathbb{I}(Y \in \mathcal{S})$ that does not reside in standard continuous kernel spaces, we introduce a smooth surrogate w_η with an approximation error bound to the indicator. Although surrogate smoothing and action-dependent evaluation are separate challenges, they can be *unified* under the RKHS representation when the action-dependent component is captured via CMEs of identifiable conditional distributions from observational data. Building on this formulation, we establish identifiability of the smoothed AUF objective and develop a kernel ridge regression (KRR) based nested estimator using historical observational samples with established statistical consistency. Finally, we propose a practical multi-start projected gradient ascent procedure to optimize the estimator and obtain AUF decisions. Our contributions can be summarized as:

1. We propose the first non-parametric rehearsal learning approach for AUF problem, eliminating the restrictive parametric assumptions imposed by existing methods.
2. We present a Probit surrogate to handle the discontinuity of the AUF objective and, through a kernel inner-product formulation, develop a KRR-based nested estimator for evaluating the AUF objective under candidate actions.
3. We establish theoretical error bounds and consistency guarantees for the proposed estimator, and empirically validate its effectiveness across a range of scenarios.

Related work. Rehearsal learning exploits influence relations among variables to support decision-making that optimizes the AUF probability (Zhou, 2022). Existing rehearsal approaches adopt parametric formulations, relying on assumptions such as linear systems or Gaussian noise to derive probabilistic constraints (Qin et al., 2023; Du et al., 2024; Tao et al., 2025) or tractable objectives (Du et al., 2025). Qin et al. (2025) extend the parametric paradigm to nonlinear settings using conditional normalizing flows (Winkler et al., 2019; Ardizzone et al., 2019) to estimate generation parameters, but replace the AUF objective with heuristic surrogates. While offering tractability, this method deviates from true AUF objective and often leads to suboptimal decisions (see Appx. A). In contrast, our approach directly targets AUF probability without heuristic substitutions. We also acknowledge a concurrent submission (Anonymity, 2026) (per ICML policy), which primarily focuses on order-learning; its decision-making component follows Qin et al. (2025) and thus inherits the same limitation. Due to space constraints, a broader discussion is deferred to Appx. B.

2. Background

In this section, we briefly review several key concepts.

2.1. Structural rehearsal models

The structural rehearsal model is a graphical model that characterizes influence relations for decision-making, consisting of (i) a set of (potentially time-varying) rehearsal graphs; and (ii) their associated generation equations (Qin et al., 2023). Rehearsal graph $G = (\mathbf{V}, \mathbf{E})$ captures the qualitative generating relations among variables, where each vertex corresponds to a variable in the decision task. Notably, rehearsal graphs employ bi-directional edges to connect variables that are mutually influenced. The formal definitions are provided below following Qin et al. (2023).

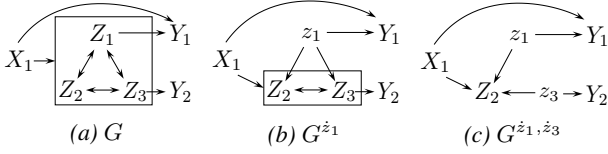


Figure 2. **2a** is an example rehearsal graph, **2b** and **2c** are graphs after rehearsal operations $\dot{z}_1$ and $\dot{z}_1, \dot{z}_3$, characterizing the data generating processes after certain alterations are performed.

Definition 1 (Mixed graph). Let $G = (\mathbf{V}, \mathbf{E})$ be a graph, where $\mathbf{V} / \mathbf{E}$ denote the vertices and the edges. G is a *mixed graph* if for any distinct vertices $u, v \in \mathbf{V}$, there is at most one edge connecting them, and the edge is either *directional* ($u \rightarrow v$ or $u \leftarrow v$) or *bi-directional* ($u \leftrightarrow v$).

Definition 2 (Bi-directional clique). A *bi-directional clique* $C = (\mathbf{V}^c, \mathbf{E}^c)$ of a mixed graph $G = (\mathbf{V}, \mathbf{E})$ is a complete subgraph induced by $\mathbf{V}^c \subseteq \mathbf{V}$ such that any edge $e \in \mathbf{E}^c$ is bi-directional. C is *maximal* if adding any other vertex does not induce a bi-directional clique.

Definition 3 (Rehearsal graph). Let $G = (\mathbf{V}, \mathbf{E})$ be a mixed graph and $\{C_i\}_{i=1}^l$ denote all maximal bi-directional cliques of G , where $C_i = (\mathbf{V}_i^c, \mathbf{E}_i^c)$. G is a *rehearsal graph* iff: (i) $\mathbf{V}_i^c \cap \mathbf{V}_j^c = \emptyset$ for any $i \neq j$; (ii) for any $i \in [l]$, if there is any edge pointing from some $u \in \mathbf{V} \setminus \mathbf{V}_i^c$ to some $v \in \mathbf{V}_i^c$, then for any $v \in \mathbf{V}_i^c$, $u \rightarrow v$; and (iii) the directional edges permit a topological ordering for $\{C_i\}_{i=1}^l$.

Associated with the graphical representation is a set of local conditional distributions, which characterizes the generating process of variables quantitatively. Given a rehearsal graph G , these distributions are defined over the set of bi-directional cliques $\{C_i\}_{i=1}^l$. Let $\mathbf{PA}_i^G \triangleq \{u \mid \exists v \in \mathbf{V}_i^c, u \rightarrow v \text{ in } G\}$ denote the parents of the clique C_i , representing external variables that influence the clique. The values of the variable sets $\mathbf{V}_i^c$ are generated based on their parents $\mathbf{PA}_i^G$ according to a general conditional distribution $P(\mathbf{V}_i^c \mid \mathbf{PA}_i^G)$ and the joint distribution $P(\mathbf{V})$ factorizes as $P(\mathbf{V}) = \prod_{i=1}^l P(\mathbf{V}_i^c \mid \mathbf{PA}_i^G)$.

To simulate the results of potential decisions, one can perform rehearsal operation $\mathbf{S} \stackrel{a}{=} \mathbf{s}$ (or abbreviated as $\dot{\mathbf{s}}$). As illustrated in Fig. 2b, applying $\dot{\mathbf{s}}$ performs a “graph surgery”: it severs all incoming edges to the decision variables $\mathbf{S}$ while assigning them specific values $\mathbf{s}$, structurally isolating the action from its natural influences. This operation blocks spurious correlations induced by confounders (e.g., U_1 in Fig. 1, which simultaneously influences both A_2 and Y_2), necessitating a sharp distinction between rehearsal distribution $P(\cdot \mid \dot{\mathbf{s}})$ and observational conditional $P(\cdot \mid \mathbf{s})$. Notation $\dot{\mathbf{s}}$ is analogous to the classic intervention operator $\hat{\mathbf{s}}$ in graphical models (Pearl, 1995). We employ the dot accent to adhere to rehearsal learning conventions accounting for bi-directed dependencies, and to strictly distinguish it from the hat notation $\hat{\cdot}$ typically reserved for statistical estimators.

2.2. Conditional mean embeddings

Let U and V be random variables in measurable domains $\mathcal{U}$ and $\mathcal{V}$, endowed with kernels $k_u : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$ and $k_v : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$, along with their associated RKHSs, denoted by $\mathcal{H}_{k_u}$ and $\mathcal{H}_{k_v}$. We assume the kernels are continuous, bounded, and positive-definite throughout the paper. The conditional mean embedding (CME) (Song et al., 2009; 2013) maps the conditional distribution $P(V|u)$ to an element $\mu_{V|u} \in \mathcal{H}_{k_v}$. Formally, it is defined as the expected feature map of V conditioned on u :

$$\mu_{V|u} := \mathbb{E}_{V \sim P(\cdot|u)}[k_v(v, \cdot)] = \int_{\mathcal{V}} k_v(v, \cdot) dP(v|u).$$

A crucial property of CME is that it enables the evaluation of conditional expectations for any function $g \in \mathcal{H}_{k_v}$ via the inner product in $\mathcal{H}_{k_v}$ that $\mathbb{E}[g(V) | u] = \langle g, \mu_{V|u} \rangle_{\mathcal{H}_{k_v}}$ (reproducing property), bridging the integral operation of expectation with the algebraic operation in the Hilbert space.

3. Our approach

In this section, we present our proposed approach for the AUF problem. We first formulate the AUF objective and introduce a smooth surrogate to handle its discontinuous. Subsequently, we propose a kernel ridge regression (KRR) based nested estimator to learn this objective from observational training data and develop a gradient-based optimization algorithm. Finally, we provide theoretical analysis establishing approximation guarantees of the smooth surrogate and the convergence rate of the proposed estimator.

3.1. Formulation of the AUF problem

Given historical data $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{z}_i, \mathbf{y}_i)\}_{i=1}^N$ (collected from observational distribution, distinct from the alterational one), we focus on the decision moment where the context $\mathbf{x}$ has just been observed, while the subsequent intermediate variables $\mathbf{Z}$ and outcomes $\mathbf{Y}$ are yet to materialize. Our objective is to determine the optimal values for the actionable subset $\mathbf{A} \subseteq \mathbf{Z}$ that maximize the probability of $\mathbf{Y}$ falling into a target desired region $\mathcal{S}$ with rehearsal operation $\dot{\mathbf{a}}$:

$$\begin{aligned} \arg \max_{\mathbf{a}} \quad & \mathbb{P}(\mathbf{Y} \in \mathcal{S} \mid \mathbf{X} = \mathbf{x}, \mathbf{A} \stackrel{a}{=} \mathbf{a}) \\ \text{s.t.} \quad & \mathbf{a}_{\text{left}} \preceq \mathbf{a} \preceq \mathbf{a}_{\text{right}}, \end{aligned} \quad (1)$$

where $\preceq$ denotes element-wise inequality, and $\mathbf{a}_{\text{left}}, \mathbf{a}_{\text{right}}$ are constant vectors specifying the feasible alteration range. Notably, unlike prior works that rely on parametric structural equations governed by θ , our formulation is non-parametric and does not assume specific functional forms for the generating process. The desired region $\mathcal{S}$ is available and characterized as a convex polytope defined by l linear constraints:

$$\mathcal{S} = \{\mathbf{y} \in \mathbb{R}^{d_y} \mid \mathbf{m}_k^T \mathbf{y} \leq b_k, k = 1, \dots, l\}. \quad (2)$$

3.2. Smoothed desirability function with identifiability

To facilitate the optimization of Eq. (1), we first reformulate the AUF probability as the expectation of a desirability indicator variable. Let $\mathbb{I}(\mathbf{y} \in \mathcal{S})$ be the indicator function which takes the value 1 if $\mathbf{y} \in \mathcal{S}$ and 0 otherwise. The objective function is equivalently expressed as:

$$\mathbb{P}(\mathbf{Y} \in \mathcal{S} \mid \mathbf{x}, \mathbf{a}) = \mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \mathbf{a})} [\mathbb{I}(\mathbf{y} \in \mathcal{S})].$$

Directly maximizing this expectation presents two fundamental challenges regarding optimization and estimation: (i) The indicator function $\mathbb{I}(\cdot)$ is discontinuous and has zero gradients almost everywhere. From a parametric perspective, calculating gradients w.r.t. $\mathbf{a}$ is intractable due to the lack of gradient signals from the objective itself. From a non-parametric kernel perspective, strictly discontinuous functions do not reside in the RKHS associated with common continuous kernels (e.g., Gaussian RBF), rendering standard kernel methods inapplicable for direct estimation. (ii) The objective depends on the alterational distribution $P(\mathbf{Y} \mid \mathbf{x}, \mathbf{a})$, which describes the outcome distribution after applying a rehearsal action. This distribution generally differs from the observational conditional $P(\mathbf{Y} \mid \mathbf{x}, \mathbf{a})$ and cannot be directly estimated from historical data $\mathcal{D}$ due to the presence of the variables influencing both $\mathbf{A}$ and $\mathbf{Y}$.

To address the first challenge, we introduce a smooth surrogate to approximate the hard indicator. Leveraging the definition of $\mathcal{S}$ in Eq. (2), we define the smoothed desirability function $w_\eta(\mathbf{y})$ as the product of Probit functions:

$$w_\eta(\mathbf{y}) \triangleq \prod_{k=1}^l \Phi(\eta(b_k - \mathbf{m}_k^\top \mathbf{y})), \quad (3)$$

where $\Phi(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution, and $\eta > 0$ is a scaling parameter controlling the sharpness of the approximation.

As illustrated in Fig. 3, $w_\eta(\mathbf{y})$ converges to $\mathbb{I}(\mathbf{y} \in \mathcal{S})$ as η increases. While the visualization depicts a univariate case, this approximation applies equally to the high-dimensional convex polytopes defined in Eq. (2). Beyond merely enabling differentiability, this surrogate offers distinct advantages. First, $w_\eta(\cdot)$ replaces the discontinuous constraints with a function that is C^∞ -continuous, ensuring numerical stability for gradient-based solvers. Second, unlike the indicator function which yields zero gradients for any $\mathbf{y} \notin \mathcal{S}$, $w_\eta(\mathbf{y})$ provides non-zero gradients even in the infeasible region. This property is particularly valuable when positive samples ($\mathbf{y} \in \mathcal{S}$) are scarce, in which case the surrogate provides an informative signal distinguishing a ‘‘near-miss’’ from a significant failure, thereby effectively guiding the optimization toward the desired region.

To address the second challenge, we establish the identifiability of the AUF objective under the smooth surrogate.

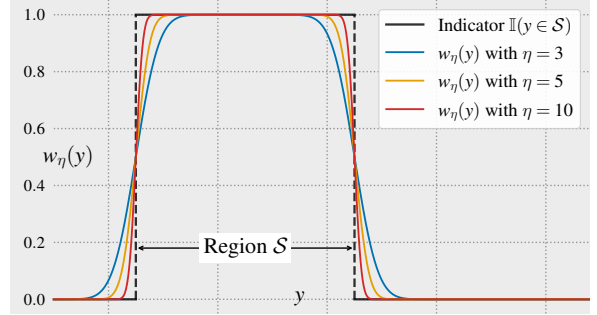


Figure 3. An example of the smooth surrogate $w_\eta(\cdot)$ approximating the hard indicator $\mathbb{I}(\cdot \in \mathcal{S})$ with varying scaling parameters η .

Definition 4 (Decomposition of $\mathbf{Z}$). Let $\mathcal{C}$ be the set of maximal bi-directional cliques in rehearsal graph G . We assume the actionable set $\mathbf{A} \subseteq \mathbf{Z}$ is a union of disjoint cliques from $\mathcal{C}$; and partition $\mathbf{Z} = \mathbf{U} \cup \mathbf{A} \cup \mathbf{U}^{\text{post}}$, where $\mathbf{U}$ denotes the *pre-alteration covariates*, consisting of variables in cliques that are ancestors of any clique in $\mathbf{A}$ according to the topological ordering of $\mathcal{C}$. Consequently, $\mathbf{U}^{\text{post}} := \mathbf{Z} \setminus (\mathbf{U} \cup \mathbf{A})$ contains the remaining post-alteration variables.

Proposition 1 (Identifiability of the AUF Objective). *Let $\mathbf{Z} = \mathbf{U} \cup \mathbf{A} \cup \mathbf{U}^{\text{post}}$ be the decomposition defined in Def. 4. The AUF objective defined by the smooth desirability approximation $w_\eta(\cdot)$ under the rehearsal operation $\mathbf{a}$ is identifiable from the observational distribution as:*

$$\mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \mathbf{a})} [w_\eta(\mathbf{Y})] = \mathbb{E}_{\mathbf{u} \sim P(\cdot \mid \mathbf{x})} [\mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \mathbf{u}, \mathbf{a})} [w_\eta(\mathbf{Y})]].$$

Prop. 1 transforms the rehearsal objective into a nested expectation over observational distributions. To streamline the notation for the subsequent estimation, we formally define the smoothed AUF objective as $J_\eta(\mathbf{a}; \mathbf{x}) \triangleq \mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \mathbf{a})} [w_\eta(\mathbf{Y})]$ and let $\mathbf{h} \triangleq (\mathbf{x}, \mathbf{u}, \mathbf{a})$ denote the concatenated vector comprising the observed context, pre-alteration covariates, and the action variables. Accordingly, we define a kernel function $k_h(\cdot, \cdot)$ over the domain of $\mathbf{h}$.

3.3. Surrogate AUF objective via nested KRR

Based on the identifiability result in Prop. 1, the AUF objective can be structured as an outer conditional expectation (over pre-alteration covariates $\mathbf{u}$) of an inner conditional expectation (over outcome $\mathbf{y}$). Hence, the evaluation naturally decomposes into a two-stage estimation problem. Let $\psi(\mathbf{h}) \triangleq \mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{h})} [w_\eta(\mathbf{Y})]$ denote the inner expectation given $\mathbf{h} = (\mathbf{x}, \mathbf{u}, \mathbf{a})$. With observed context $\mathbf{x}$ and specific alteration $\mathbf{a}$, $\psi(\mathbf{h})$ serves as the response variable dependent on $\mathbf{u}$, acting as the target for the outer expectation $\mathbb{E}_{\mathbf{u} \sim P(\cdot \mid \mathbf{x})}$. This structural dependency naturally motivates our nested approach: we first estimate $\psi(\mathbf{h})$ in outcome RKHS $\mathcal{H}_{k_y}$, and subsequently use these estimates to solve the outer expectation in context RKHS $\mathcal{H}_{k_x}$.

Stage 1: Inner Expectation Estimation. We begin by deriving the estimator for the inner component $\psi(\mathbf{h})$. While the original desirability indicator $\mathbb{I}(\mathbf{y} \in \mathcal{S})$ is discontinuous and generally does not reside in the RKHS associated with continuous kernels, the proposed smooth surrogate w_η possesses sufficient regularity. By assuming that w_η resides in the RKHS associated with kernel k_y , i.e., $w_\eta \in \mathcal{H}_{k_y}$, we can leverage the *reproducing property* $w_\eta(\mathbf{y}) = \langle w_\eta, k_y(\cdot, \mathbf{y}) \rangle_{\mathcal{H}_{k_y}}$ to reformulate the inner expectation. Let $\mu_{\mathbf{Y}|\mathbf{h}} \in \mathcal{H}_{k_y}$ denote the CME of the distribution $P(\mathbf{y} | \mathbf{h})$. The expectation can be derived as:

$$\begin{aligned} \mathbb{E}_{\mathbf{y} \sim P(\cdot|\mathbf{h})}[w_\eta(\mathbf{Y})] &= \int_{\mathcal{Y}} \langle w_\eta, k_y(\cdot, \mathbf{y}) \rangle_{\mathcal{H}_{k_y}} p(\mathbf{y} | \mathbf{h}) d\mathbf{y} \\ &= \left\langle w_\eta, \int_{\mathcal{Y}} k_y(\cdot, \mathbf{y}) p(\mathbf{y} | \mathbf{h}) d\mathbf{y} \right\rangle_{\mathcal{H}_{k_y}} \\ &= \langle w_\eta, \mu_{\mathbf{Y}|\mathbf{h}} \rangle_{\mathcal{H}_{k_y}}. \end{aligned} \quad (4)$$

Consequently, the evaluation of the expectation hinges on the accurate estimation of CME $\mu_{\mathbf{Y}|\mathbf{h}}$. To address this, we employ kernel ridge regression (KRR), a non-parametric framework that effectively captures complex non-linear dependencies in the RKHS while admitting a computationally efficient closed-form solution. Given the observational dataset $\mathcal{D} = \{(\mathbf{h}_i, \mathbf{y}_i)\}_{i=1}^N$, where $\mathbf{h}_i$ collects the historical covariates and $\mathbf{y}_i$ denotes the outcome, the CME $\mu_{\mathbf{Y}|\mathbf{h}}$ can be empirically estimated via KRR as:

$$\begin{aligned} \hat{\mu}_{\mathbf{Y}|\mathbf{h}} &= \sum_{i=1}^N \beta_i(\mathbf{h}) k_y(\cdot, \mathbf{y}_i), \\ \text{with } \beta(\mathbf{h}) &= (\mathbf{K}_{hh} + N\lambda_h \mathbf{I})^{-1} \mathbf{k}_h(\mathbf{h}), \end{aligned} \quad (5)$$

where $\mathbf{k}_h(\mathbf{h}) = [k_h(\mathbf{h}_1, \mathbf{h}), \dots, k_h(\mathbf{h}_N, \mathbf{h})]^\top$, $\lambda_h > 0$ is the regularization parameter, and $\mathbf{K}_{hh} \in \mathbb{R}^{N \times N}$ is the Gram matrix with entries $[\mathbf{K}_{hh}]_{ij} = k_h(\mathbf{h}_i, \mathbf{h}_j)$. Substituting the estimated CME from Eq. (5) back into the inner product formulation in Eq. (4) and let $\mathbf{w}_\eta = [w_\eta(\mathbf{y}_1), \dots, w_\eta(\mathbf{y}_N)]^\top$, it can be derived that:

$$\begin{aligned} \hat{\mathbb{E}}_{\mathbf{y} \sim P(\cdot|\mathbf{h})}[w_\eta(\mathbf{Y})] &= \left\langle w_\eta, \sum_{i=1}^N \beta_i(\mathbf{h}) k_y(\cdot, \mathbf{y}_i) \right\rangle_{\mathcal{H}_{k_y}} \\ &= \sum_{i=1}^N \beta_i(\mathbf{h}) \langle w_\eta, k_y(\cdot, \mathbf{y}_i) \rangle_{\mathcal{H}_{k_y}} \\ &= \mathbf{w}_\eta^\top (\mathbf{K}_{hh} + N\lambda_h \mathbf{I})^{-1} \mathbf{k}_h(\mathbf{h}). \end{aligned} \quad (6)$$

Stage 2: Outer Expectation Estimation. We now proceed to the outer expectation over the pre-alteration covariates $\mathbf{u}$. Note that the term $\alpha^\top \triangleq \mathbf{w}_\eta^\top (\mathbf{K}_{hh} + N\lambda_h \mathbf{I})^{-1}$ in Eq. (6) is a constant row vector determined solely by the observational training data. To facilitate the analytical integration over $\mathbf{u}$, we introduce Ass. 1, a mild structural condition on the kernel that is satisfied by a lot of standard kernels, including the Gaussian RBF and polynomial kernels.

Assumption 1 (Decomposable Kernels). We assume that the kernel $k_h(\cdot, \cdot)$ defined on the joint space $\mathbf{h} = (\mathbf{x}, \mathbf{u}, \mathbf{a})$ admits a multiplicative decomposition:

$$k_h(\mathbf{h}_i, \mathbf{h}) = k_x(\mathbf{x}_i, \mathbf{x}) \cdot k_u(\mathbf{u}_i, \mathbf{u}) \cdot k_a(\mathbf{a}_i, \mathbf{a}).$$

With this assumption, we substitute the estimator $\hat{\psi}(\mathbf{h})$ back into the outer expectation. Since the integration is performed solely w.r.t. $\mathbf{u}$, the terms of $\mathbf{x}$ and $\mathbf{a}$ remain constant within the integral, thus $\mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}(\mathbf{x}, \mathbf{u}, \mathbf{a})]$ can be computed as:

$$\begin{aligned} &\int_{\mathcal{U}} \left(\sum_{i=1}^N \alpha_i k_x(\mathbf{x}_i, \mathbf{x}) k_u(\mathbf{u}_i, \mathbf{u}) k_a(\mathbf{a}_i, \mathbf{a}) \right) p(\mathbf{u} | \mathbf{x}) d\mathbf{u} \\ &= \sum_{i=1}^N \alpha_i \underbrace{k_x(\mathbf{x}_i, \mathbf{x})}_{\text{Context Match}} \underbrace{k_a(\mathbf{a}_i, \mathbf{a})}_{\text{Action similarity}} \underbrace{\int_{\mathcal{U}} k_u(\mathbf{u}_i, \mathbf{u}) p(\mathbf{u} | \mathbf{x}) d\mathbf{u}}_{\text{Expected Unactionable Terms}}. \end{aligned} \quad (7)$$

Similar to Eq. (5), the second KRR estimator of CME $\mu_{\mathbf{U}|\mathbf{x}}$ can be given as $\hat{\mu}_{\mathbf{U}|\mathbf{x}} = \sum_{j=1}^N \gamma_j(\mathbf{x}) k_u(\cdot, \mathbf{u}_j)$, with $\gamma(\mathbf{x}) = (\mathbf{K}_{xx} + N\lambda_x \mathbf{I})^{-1} \mathbf{k}_x(\mathbf{x})$. Leveraging this estimator, we can explicitly expand $\int_{\mathcal{U}} k_u(\mathbf{u}_i, \mathbf{u}) p(\mathbf{u} | \mathbf{x}) d\mathbf{u}$ in Eq. (7) as:

$$\begin{aligned} \int_{\mathcal{U}} k_u(\mathbf{u}_i, \mathbf{u}) p(\mathbf{u} | \mathbf{x}) d\mathbf{u} &= \langle k_u(\mathbf{u}_i, \cdot), \hat{\mu}_{\mathbf{U}|\mathbf{x}} \rangle_{\mathcal{H}_{k_u}} \\ &= \left\langle k_u(\mathbf{u}_i, \cdot), \sum_{j=1}^N \gamma_j(\mathbf{x}) k_u(\cdot, \mathbf{u}_j) \right\rangle_{\mathcal{H}_{k_u}} \\ &= \sum_{j=1}^N \gamma_j(\mathbf{x}) \underbrace{\langle k_u(\mathbf{u}_i, \cdot), k_u(\cdot, \mathbf{u}_j) \rangle_{\mathcal{H}_{k_u}}}_{k_u(\mathbf{u}_i, \mathbf{u}_j)}. \end{aligned} \quad (8)$$

Let $\mathbf{K}_{uu} \in \mathbb{R}^{N \times N}$ denote the Gram matrix with entries $[\mathbf{K}_{uu}]_{ij} = k_u(\mathbf{u}_i, \mathbf{u}_j)$, Eq. (8) equals the i -th row of vector $\mathbf{K}_{uu} \gamma(\mathbf{x})$. Substituting Eq. (8) back into Eq. (7) yields the closed-form nested estimator of original AUF objective:

$$\begin{aligned} &\hat{\mathbb{E}}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\mathbb{E}}_{\mathbf{y} \sim P(\cdot|\mathbf{h})}[w_\eta(\mathbf{Y})]] \\ &= \sum_{i=1}^N \alpha_i [\mathbf{K}_{uu} \gamma(\mathbf{x})]_i k_x(\mathbf{x}_i, \mathbf{x}) k_a(\mathbf{a}_i, \mathbf{a}) \\ &= (\alpha \odot \mathbf{k}_x(\mathbf{x}) \odot (\mathbf{K}_{uu} \gamma(\mathbf{x})))^\top \mathbf{k}_a(\mathbf{a}), \end{aligned}$$

where $\odot$ denotes the Hadamard product. This is to say, let $\omega(\mathbf{x}) = (\alpha \odot \mathbf{k}_x(\mathbf{x}) \odot (\mathbf{K}_{uu} \gamma(\mathbf{x})))$, the final surrogate of AUF objective simplifies to a linear function of the alteration's feature map in the RKHS:

$$\hat{J}_\eta(\mathbf{a}; \mathbf{x}) = \omega(\mathbf{x})^\top \mathbf{k}_a(\mathbf{a}). \quad (9)$$

This formulation enables optimizing the proposed alteration $\mathbf{a}$ by maximizing its weighted alignment with the observed realizations of actionable variables in the training set. The weight vector $\omega(\mathbf{x})$ integrates the desirability of historical

outcomes (via α) with the relevance of the current context (via $\mathbf{k}_x \mathbf{x}$). Meanwhile, the term $\mathbf{K}_{uu}$ functions as a confounder adjustment bridge: it measures the similarity between expected confounder profile (via $\gamma(\mathbf{x})$) and the actual pre-alteration covariates observed in historical samples. This mechanism effectively re-weights the observational data to mitigate the distributional shift induced by actions, ensuring that the optimization targets the influence of the decision alteration $\mathbf{a}$ on the target variables $\mathbf{Y}$ given the context $\mathbf{X}$. Note that in the degenerate case where $\mathcal{U} = \emptyset$, adjustment term $\mathbf{K}_{uu}\gamma(\mathbf{x})$ simplifies to an all-one vector, reflecting that no confounder adjustment is required.

3.4. Gradient-based rehearsal learning

Having derived the closed-form surrogate $\hat{J}_\eta(\mathbf{a}; \mathbf{x})$, our goal is to find the optimal rehearsal alteration $\mathbf{a}^*$ that maximizes this objective subject to the feasible constraints:

$$\mathbf{a}^* = \arg \max_{\mathbf{a}} \hat{J}_\eta(\mathbf{a}; \mathbf{x}) \quad \text{s.t.} \quad \mathbf{a}_{\text{left}} \preceq \mathbf{a} \preceq \mathbf{a}_{\text{right}}.$$

To enable efficient gradient-based optimization and ensure consistency across the framework, we specify all kernels involved in our approach (*i.e.*, k_x, k_u, k_a, k_y) as Radial Basis Function (RBF) kernels. For a generic variable $\mathbf{v} \in \{\mathbf{x}, \mathbf{u}, \mathbf{a}, \mathbf{y}\}$, the corresponding kernel is defined as:

$$k_v(\mathbf{v}, \mathbf{v}') = \exp\left(-\frac{\|\mathbf{v} - \mathbf{v}'\|_2^2}{2\sigma_v^2}\right),$$

where σ_v denotes the bandwidth parameter specific to each variable type. This specification naturally satisfies the decomposable assumption in Ass. 1 and provides the necessary smoothness for the subsequent gradient derivation. Under this specification, the objective function becomes a weighted sum of Gaussian bumps centered at historical actions $\mathbf{a}_i$:

$$\hat{J}_\eta(\mathbf{a}; \mathbf{x}) = \sum_{i=1}^N \omega_i(\mathbf{x}) \exp\left(-\frac{\|\mathbf{a}_i - \mathbf{a}\|_2^2}{2\sigma_a^2}\right).$$

Recall that the weight $\omega_i(\mathbf{x}) = [\alpha \odot \mathbf{k}_x(\mathbf{x}) \odot (\mathbf{K}_{uu}\gamma(\mathbf{x}))]_i$ encapsulates the confounding-adjusted desirability of the i -th historical sample given the current context $\mathbf{x}$. Crucially, unlike coefficients in standard density estimation, $\omega_i(\mathbf{x})$ can take negative values as it stems from the analytical KRR solutions involving matrix inversions.

Since the surrogate objective is differentiable, we can derive the gradient with respect to $\mathbf{a}$ analytically. By applying the chain rule, the gradient is given by:

$$\begin{aligned} \nabla_{\mathbf{a}} \hat{J}_\eta(\mathbf{a}; \mathbf{x}) &= \sum_{i=1}^N \omega_i(\mathbf{x}) \nabla_{\mathbf{a}} k_a(\mathbf{a}_i, \mathbf{a}) \\ &= \frac{1}{\sigma_a^2} \sum_{i=1}^N [\omega_i(\mathbf{x}) k_a(\mathbf{a}_i, \mathbf{a})] (\mathbf{a}_i - \mathbf{a}). \end{aligned} \quad (10)$$

Algorithm 1 Rehearsal Learning via Nested Estimator

```

1: Input: Observational data  $\mathcal{D}$ , context  $\mathbf{x}$ , region  $\mathcal{S}$ .
2: Hyperparameters: Kernel params  $\{\sigma_h, \sigma_x, \sigma_u, \sigma_a\}$ ,
   regularization  $\{\lambda_h, \lambda_x\}$ , smoothing  $\eta$ , learning rate  $\nu$ .
3: // Phase 1: Offline Learning (One-time)
4: Compute Gram matrices  $\mathbf{K}_{hh}, \mathbf{K}_{xx}, \mathbf{K}_{uu}$ .
5: Compute smoothed desirability vector  $\mathbf{w}_\eta$  using Eq. (3).
6: Solve stage 1 coeffs:  $\alpha^\top \leftarrow \mathbf{w}_\eta^\top (\mathbf{K}_{hh} + N\lambda_h \mathbf{I})^{-1}$ .
7: Solve stage 2 coeffs:  $\mathbf{G} \leftarrow (\mathbf{K}_{xx} + N\lambda_x \mathbf{I})^{-1}$ .
8: // Phase 2: Rehearsal Operation (Per context  $\mathbf{x}$ )
9: Compute kernel vectors  $\mathbf{k}_x(\mathbf{x})$ .
10:  $\gamma(\mathbf{x}) \leftarrow \mathbf{G} \mathbf{k}_x(\mathbf{x})$ .
11:  $\omega(\mathbf{x}) \leftarrow \alpha \odot \mathbf{k}_x(\mathbf{x}) \odot (\mathbf{K}_{uu}\gamma(\mathbf{x}))$ .
12: // Phase 3: Multi-start Optimization
13:  $\mathcal{A}_{\text{init}} \leftarrow$  Top- $K$  indices  $i$  where  $\omega_i(\mathbf{x}) > 0$ .
14:  $\mathcal{S}_{\text{sols}} \leftarrow \emptyset$ 
15: for each  $i \in \mathcal{A}_{\text{init}}$  do
16:   Initialize  $\mathbf{a}^{(0)} \leftarrow \mathbf{a}_i$ .
17:   for  $t = 1$  to  $T$  do
18:     Compute gradient  $\nabla_{\mathbf{a}} \hat{J}_\eta(\mathbf{a}^{(t-1)}; \mathbf{x})$  via Eq. (10).
19:     Update:  $\mathbf{a}' \leftarrow \mathbf{a}^{(t-1)} + \nu \nabla_{\mathbf{a}} \hat{J}_\eta$ .
20:     Project:  $\mathbf{a}^{(t)} \leftarrow \mathcal{P}_{\{\mathbf{a} | \mathbf{a}_{\text{left}} \preceq \mathbf{a} \preceq \mathbf{a}_{\text{right}}\}}(\mathbf{a}')$ .
21:   end for
22:   Store solution:  $\mathcal{S}_{\text{sols}} \leftarrow \mathcal{S}_{\text{sols}} \cup \{\mathbf{a}^{(T)}\}$ .
23: end for
24: Output:  $\mathbf{a}^* = \arg \max_{\mathbf{a} \in \mathcal{S}_{\text{sols}}} \hat{J}_\eta(\mathbf{a}; \mathbf{x})$ .
    
```

This gradient interpretation is intuitive: it acts as a force vector that pulls the candidate $\mathbf{a}$ towards historical actions $\mathbf{a}_i$ that have high desirability weights $\omega_i(\mathbf{x})$, while being repelled by those with negative weights. Meanwhile, objective $\hat{J}_\eta(\mathbf{a}; \mathbf{x})$ presents a potentially non-convex landscape since structured as a sum of Gaussians with mixed-sign weights $\omega_i(\mathbf{x})$, thus random initialization strategy risks entrapment in poor local optima. To maximize the estimated objective, we employ a multi-start strategy initialized at historical actions that exhibit high estimated utility. Specifically, we construct the candidate set $\mathcal{A}_{\text{init}}$ by selecting historical $\mathbf{a}_i$ whose corresponding weight values $\omega_i(\mathbf{x})$ are among the top- K (up to total number of) positive entries in $\omega(\mathbf{x})$:

$$\mathcal{A}_{\text{init}} = \{\mathbf{a}_i \in \mathcal{D} \mid \omega_i(\mathbf{x}) > 0 \text{ and is among top-}K\}.$$

Alg. 1 outlines the full decision-making procedure. It starts with an offline stage where kernel Gram matrices and corresponding KRR coefficients are computed from historical data. Given a newly observed context $\mathbf{x}$, the algorithm then computes the context-dependent mixing weights $\omega(\mathbf{x})$, which are subsequently used to initialize a multi-start projected gradient ascent procedure that iteratively performs gradient ascent and projects the updates onto the feasible set, eventually yielding the optimal rehearsal alteration $\mathbf{a}^*$. The overall time complexity is $\mathcal{O}(N^3)$, dominated by the matrix inversion required in the offline stage.

3.5. Theoretical analysis

We first establish the approximation bound for the smooth desirability function as follows.

Theorem 1. *Let $\mathcal{S}$ be the convex polytope region defined in Eq. (2), and let $w_\eta(\mathbf{y})$ be the Probit surrogate of the indicator function $\mathbb{I}(\mathbf{y} \in \mathcal{S})$ as defined in Eq. (3). Assuming the conditional density is bounded, the gap between the true AUF probability and the surrogate objective satisfies:*

$$\left| \mathbb{P}(\mathbf{Y} \in \mathcal{S} \mid \mathbf{x}, \mathbf{a}) - \mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \mathbf{a})} [w_\eta(\mathbf{y})] \right| = \mathcal{O} \left(\frac{\sqrt{\ln \eta}}{\eta} \right).$$

Thm. 1 provides a theoretical justification for utilizing the Probit surrogate. The established bound guarantees that the discrepancy is tightly controlled by the scaling parameter η , vanishing at a rate of $\tilde{\mathcal{O}}(1/\eta)$ as $\eta \rightarrow \infty$. While a larger η yields a tighter approximation of the indicator function, it is worth noting that η cannot be arbitrarily large in practice. As $\eta \rightarrow \infty$, the surrogate $w_\eta(\cdot)$ approaches a discontinuity, causing its RKHS norm to grow significantly, in which case an excessively large η would require a prohibitive number of observational samples to estimate accurately. Thus, η serves as a trade-off parameter balancing the approximation bias and the learnability (estimation variance) of the objective. Next, we characterize the statistical performance of our proposed nested estimator $\hat{J}_\eta(\mathbf{a}; \mathbf{x})$ given a fixed η .

Theorem 2. *Let N be the sample size of the observational training data, and let $\lambda_h, \lambda_x > 0$ denote the regularization parameters for the inner (Stage 1) and outer (Stage 2) KRR estimators. Suppose that w_η is bounded in the RKHS norm, i.e., $\|w_\eta\|_{\mathcal{H}_{k_y}} \leq C_w$. For any context $\mathbf{x}$ and alteration $\mathbf{a}$, the estimation error $\Delta_{est} \triangleq \left| \hat{J}_\eta(\mathbf{a}; \mathbf{x}) - J_\eta(\mathbf{a}; \mathbf{x}) \right|$ satisfies:*

$$\Delta_{est} = \mathcal{O}_p \left(\sqrt{\lambda_x} + \sqrt{\lambda_h} + \frac{1}{\sqrt{N\lambda_x}} + \frac{1}{\sqrt{N\lambda_h^3}} \right).$$

Thm. 2 explicitly characterizes the learning speed of our nested estimator. This bound implies the consistency of the proposed method: provided that the regularization parameters decay appropriately (i.e., $\lambda_x, \lambda_h \rightarrow 0$) while satisfying $N\lambda_x \rightarrow \infty$ and $N\lambda_h^3 \rightarrow \infty$ as $N \rightarrow \infty$, the estimation error converges to zero in probability. For instance, a valid configuration satisfying these conditions is $\lambda_x = \mathcal{O}(N^{-1/2})$ and $\lambda_h = \mathcal{O}(N^{-1/4})$. With appropriate assumptions on the joint distribution of $\mathbf{X}$, $\mathbf{U}$, $\mathbf{A}$ and $\mathbf{Y}$, better rates can be obtained (Grünwälder et al., 2012a).

Finally, since Thm. 1 characterizes the approximation error of the smoothed surrogate objective and Thm. 2 bounds the estimation error induced by finite samples, these results together establish that, with appropriately chosen smoothing hyperparameters and regularization hyperparameters, the established learnable estimator $\hat{J}_\eta$ provides a reliable proxy for the original intractable AUF objective.

4. Experiments

In this section, we present the experimental results to validate the effectiveness of our proposed method. All experiments were conducted on an NVIDIA Tesla A100 GPU and two Intel Xeon Platinum 8358 CPUs, while specific experimental details are listed in Appx. D. We first briefly introduce the experimental datasets as follows.

Linear setting. We utilize the Bermuda dataset (Courtney et al., 2017; Andersson & Bates, 2018) that records environmental variables in the Bermuda area, with the objective of maintaining a high net coral ecosystem calcification (NEC) level. The data-generating graph structure is known (Courtney et al., 2017), and the parameters are derived by fitting linear models. The dimensions of $\mathbf{X}$, $\mathbf{Z}$, and $\mathbf{Y}$ are 3, 7, and 1, respectively, and the desired region is defined as $\mathcal{S} = \{\text{NEC} \in [0.5, 2]\}$. Additionally, we employ the synthetic dataset consistent with Du et al. (2025), where the dimensions of $\mathbf{X}$, $\mathbf{Z}$, and $\mathbf{Y}$ are 2, 4, and 2, respectively, and the desired region $\mathcal{S}$ for $\mathbf{Y}$ is a rectangle area.

Nonlinear setting. We implement the motivating loan approval example illustrated in Fig. 1. To facilitate visualization, we focus on altering solely on the interest rate (A_2). The objective is to ensure a high repayment rate (Y_1) while maintaining a favorable return on ROI (Y_2), thus the desired region is defined as $\mathcal{S} = \{(Y_1, Y_2) \mid Y_1 \geq 0.6, Y_2 \geq 0.3\}$. Furthermore, to evaluate performance under complex conditions, we implemented several synthetic datasets featuring non-linear mechanisms (including square, trigonometric, and Sigmoid transformations) and non-Gaussian randomness (including uniform, beta, and exponential distributions). Further details are available in Appx. D.

We evaluate our method against rehearsal learning baselines, including QWZ23 (Qin et al., 2023), MICNS (Du et al., 2024), CARE (Du et al., 2025), and Grad-Rh (Qin et al., 2025). It is worth noting that CARE is theoretically optimal for linear additive Gaussian data, and Grad-Rh is a heuristic approach compatible with non-linear scenarios. For all configurations, the initial observational dataset size is set to $|\mathcal{D}| = 1,000$, and we report the average performance under 5 different random seeds. In our proposed Alg. 1, optimization is initialized with $K = 20$ starting points and a learning rate of 5^{-1} . Kernel bandwidths are determined using the median heuristic (Gretton et al., 2012) with minor dataset-specific fine-tuning. The smoothing parameter η for the indicator approximation is adaptively selected from $\{5, 10, 20\}$ based on the proportion of positive samples (i.e., $\mathbf{y}_i \in \mathcal{S}$) in $\mathcal{D}$; a larger η is employed when positive samples are abundant to allow for sharper approximation. Lastly, the decision quality is evaluated as the probability of successfully avoiding the undesired future, calculated via the empirical success frequency over 100 Monte Carlo trials sampled from the ground-truth rehearsal distribution.

Table 1. The AUF probability $\mathbb{P}(\mathbf{Y} \in \mathcal{S} \mid \mathbf{x}, \hat{\mathbf{a}})$ evaluated on five datasets. Each value is estimated using 100 Monte Carlo samples, averaged over 5 random seeds. Results are reported as mean $\pm$ standard deviation. The best results are highlighted in bold.

Dataset	Baseline (No action)	QWZ23	MICNS	CARE	Grad-Rh	Ours
Bermuda	0.222 $\pm$ 0.032	0.676 $\pm$ 0.027	0.668 $\pm$ 0.079	0.706 $\pm$ 0.065	0.674 $\pm$ 0.030	0.702 $\pm$ 0.062
Lin-Syn1	0.166 $\pm$ 0.160	0.938 $\pm$ 0.032	0.912 $\pm$ 0.038	0.956 $\pm$ 0.019	0.934 $\pm$ 0.022	0.942 $\pm$ 0.016
BankExp	0.598 $\pm$ 0.066	0.563 $\pm$ 0.019	0.576 $\pm$ 0.014	0.640 $\pm$ 0.054	0.698 $\pm$ 0.055	0.820 $\pm$ 0.078
Non-Syn1	0.064 $\pm$ 0.014	0.130 $\pm$ 0.083	0.114 $\pm$ 0.091	0.078 $\pm$ 0.058	0.291 $\pm$ 0.107	0.430 $\pm$ 0.061
Non-Syn2	0.186 $\pm$ 0.068	0.320 $\pm$ 0.237	0.324 $\pm$ 0.196	0.392 $\pm$ 0.244	0.362 $\pm$ 0.243	0.584 $\pm$ 0.086

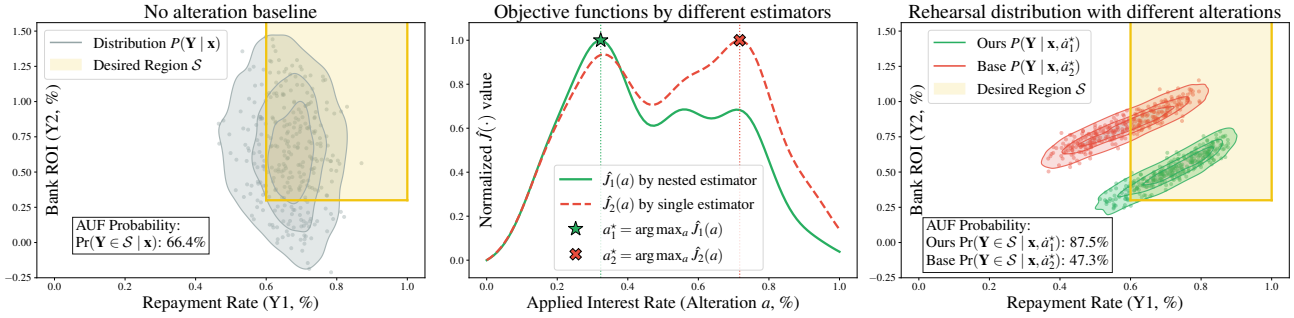


Figure 4. Illustrative example of loan interest rate optimization as shown in Fig. 1. The left panel shows the distribution conditioned on context $\mathbf{x}$ without active alterations. The middle panel displays the comparison of objective functions estimated by single estimator and nested estimator. The right panel illustrates the rehearsal distributions under the alterations optimized by the two estimators.

Experimental results are presented in Tab. 1. Baseline results denote the natural AUF probability, *i.e.*, $\mathbb{P}(\mathbf{Y} \in \mathcal{S} \mid \mathbf{x})$, observed without any alterations. In the linear additive Gaussian settings, our method outperforms all rehearsal learning baselines except CARE. This performance gap is expected, as CARE is theoretically proven to be optimal under these specific conditions. However, it is crucial to highlight that CARE (and most other baselines, excluding Grad-Rh) know the true parametric family in this case, in contrast, our approach is non-parametric thus does not use prior knowledge of the underlying functional forms. Consequently, in the non-linear settings where the specific parametric assumptions of prior methods are violated, our method demonstrates superior adaptability and robustness, outperforming all comparison methods.

To illustrate the necessity of optimizing the rehearsal objective $\mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \hat{\mathbf{a}})}[w_\eta(\mathbf{Y})]$ (identifiable as in Prop. 1), rather than directly maximizing $\mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \mathbf{a})}[w_\eta(\mathbf{Y})]$ (pure conditional), we present a comparative analysis on the lending task in Fig. 1. Initially, distribution $P(\mathbf{Y} \mid \mathbf{x})$ shows a baseline probability of 66.4% within the desired region $\mathcal{S}$ without active alterations. Driven by the goal to actively perform rehearsal operations to increase the probability of outcomes falling into $\mathcal{S}$, we compare optimal alterations suggested by maximizing the two estimators. Maintaining consistent hyperparameters (*e.g.*, kernel bandwidth), the landscape estimated by single estimator (constructed to estimate $\mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \mathbf{a})}[w_\eta(\mathbf{Y})]$) is misled by the unactionable industry stability U that influences both $\mathbf{A}$ and $\mathbf{Y}$, suggest-

ing a suboptimal action a_2^* . In contrast, our nested estimator explicitly eliminates this confounding, correctly identifying the true optimal alteration a_1^* . Consequently, the resulting rehearsal distributions confirm that the misled alteration yields a detrimental effect (AUF probability 47.3%), performing even worse than the no-alteration baseline. Conversely, our method successfully shifts outcomes into $\mathcal{S}$ with an AUF probability of 87.5%, demonstrating the critical necessity of modeling $\mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \hat{\mathbf{a}})}[w_\eta(\mathbf{Y})]$ by our nested estimator.

5. Conclusion

In this paper, we presented the first non-parametric rehearsal learning approach for the AUF problem, addressing the key limitations of existing rehearsal-based methods that rely on restrictive parametric assumptions. By avoiding specific functional forms of the data generation process, our approach extends rehearsal learning to complex, nonlinear real-world systems. Methodologically, we introduced a smooth Probit approximation to handle the discontinuous desirability indicator and developed a KRR-based nested estimator for the surrogate AUF objective. This estimator enables direct and effective learning of the AUF objective from observational data, grounded in the theory of conditional mean embeddings. Theoretically, we established identifiability of the AUF objective and derived error bounds for both the smooth approximation and statistical estimation, guaranteeing consistency of the proposed approach. Empirical results demonstrate the effectiveness and flexibility.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Discussion

Prior rehearsal learning methodologies have been predominantly confined to the assumption that the underlying generation mechanisms follow linear additive noise models. While [Qin et al. \(2025\)](#) propose a flow-based rehearsal learning method purportedly adaptable to nonlinear systems, a critical limitation lies in its reliance on a heuristic surrogate objective. To circumvent the intractability of directly optimizing the AUF probability, [Qin et al. \(2025\)](#) propose identifying the alteration $\mathbf{a}$ that minimizes $\|\mu_{\mathbf{Y}|\mathbf{x},\mathbf{a}} - \mathbf{c}\|_2$, where $\mathbf{c}$ denotes the centroid of region $\mathcal{S}$. However, this heuristic often misaligns with true AUF objective (maximizing $\mathbb{P}(\mathbf{Y} \in \mathcal{S} | \mathbf{x}, \mathbf{a})$). As illustrated in Fig. 5 ($\mathcal{S}$ is an equilateral triangle, a common polytope), the strategy minimizing the heuristic loss (blue) perfectly aligns the distribution mean with the centroid ($\|\mu_1 - \mathbf{c}\| = 0$) yet only yields a success probability of 0.577. Conversely, an alternative strategy (green) explicitly violates the heuristic condition ($\|\mu_2 - \mathbf{c}\| > 0$) but achieves a significantly higher probability of 0.709. This demonstrates that this distance-based surrogate can result in suboptimal decisions, diverging from the goal of AUF.

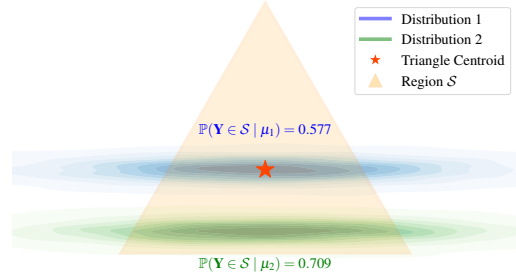


Figure 5. A counter-example showing the misalignment between heuristic distance minimization and true AUF objective.

To surmount this limitation, we leverage CMEs to reconstruct the optimization problem. We begin by approximating the discontinuous indicator function $\mathbb{I}(\cdot)$ with a continuous one $w_\eta(\cdot)$, allowing us to reformulate the maximization of $\mathbb{P}(\mathbf{Y} \in \mathcal{S} | \mathbf{x}, \mathbf{a})$ into the maximization of the inner product between w_η and the CME of $\mathbf{Y}$ within the RKHS. Subsequently, we nestedly utilize KRR to estimate the CME, yielding a kernel-based surrogate AUF objective. Theoretically, Thm. 1 and Thm. 2 guarantee that the deviation between our surrogate and the true AUF objective is bounded, ensuring optimization consistency. A significant challenge inherent in our non-parametric approach is hyperparameter selection, including η for smooth approximation, bandwidths of RBF kernels, and regularization parameters in KRR. In this work, hyperparameters were determined empirically, referencing established heuristics in kernel literature ([Gretton et al., 2012](#)). A critical avenue for future research is to explore principled methods for hyperparameter selection in non-parametric rehearsal learning.

B. Additional related work

We review two types of decision-related methods, including reinforcement learning (RL) and causal methods:

Reinforcement Learning. RL has demonstrated strong empirical performance across various decision-making problems ([Sutton & Barto, 2018](#)). Conventional RL algorithms ([Schulman et al., 2017](#); [Haarnoja et al., 2018](#)) typically depend on repeated and potentially costly interactions with the environment, rendering them unsuitable for AUF settings where such interactions are expensive and limited ([Zhou, 2022](#); [Qin et al., 2023](#)). Recent progress in offline and hybrid offline–online RL ([Song et al., 2023](#); [Pong et al., 2022](#); [Li et al., 2023](#); [Qiao & Wang, 2023](#)) seeks to alleviate this issue by exploiting pre-collected data; however, hybrid approaches presume the feasibility of online interaction for policy refinement, which remains inapplicable to high-stakes AUF scenarios and thus cannot handle the confounding bias inherent in passive observational data. Besides, several model-based RL approaches aim to mitigate confounding effects in offline policy evaluation ([Kallus & Zhou, 2020](#); [Bruns-Smith, 2021](#); [Kausik et al., 2024](#)). In contrast to our framework, which explicitly utilizes the full graphical structure without imposing strong assumptions on the confounding, these approaches often rely on restrictive structural conditions and commonly assume that confounders influence only the transition dynamics.

Causality. A substantial body of work has investigated the use of structural models for decision-making, with most approaches rooted in the framework of structural causal models (SCMs) ([Spirtes et al., 2000](#)). Several studies focus on system identification ([He & Geng, 2008](#); [Zhang & Hyvärinen, 2009](#); [Kocaoglu et al., 2017](#); [Zhang et al., 2017](#); [Cai et al., 2018](#); [Shiragur et al., 2024](#)). However, these methods typically emphasize recovering the underlying causal structure itself, rather than addressing downstream objectives such as estimating or optimizing decision-relevant causal effects. In parallel, causal bandit frameworks have been proposed to tackle optimal arm identification problems ([Bareinboim et al., 2015](#); [Lattimore et al., 2016](#); [Sen et al., 2017](#); [Lee & Bareinboim, 2018](#); [Zhang & Bareinboim, 2019](#); [Lee & Bareinboim, 2020](#); [Lu et al., 2021](#); [Park et al., 2025](#); [Park & Lee, 2025](#)). Such approaches generally seek a single, globally optimal action that maximizes the expected reward. By contrast, rehearsal learning methods naturally accommodates mutually influenced relationships among variables and targets the maximization of the AUF probability by selecting context-dependent optimal alterations conditioned on the observed context $\mathbf{x}$, and allows for a more expressive specification of the desired outcome region $\mathcal{S}$, extending beyond simple expectation maximization objectives.

C. Proofs

In this section, we provide proof for claims in the main text.

C.1. Proof of Proposition 1

Proposition 1. *Let $\mathbf{Z} = \mathbf{U} \cup \mathbf{A} \cup \mathbf{U}^{post}$ be the decomposition defined in Def. 4. The AUF objective defined by the smooth desirability approximation $w_\eta(\cdot)$ under the rehearsal operation $\dot{\mathbf{a}}$ is identifiable from the observational distribution as:*

$$\mathbb{E}_{\mathbf{y} \sim P(\cdot | \mathbf{x}, \dot{\mathbf{a}})}[w_\eta(\mathbf{Y})] = \mathbb{E}_{\mathbf{u} \sim P(\cdot | \mathbf{x})}[\mathbb{E}_{\mathbf{y} \sim P(\cdot | \mathbf{x}, \mathbf{u}, \mathbf{a})}[w_\eta(\mathbf{Y})]] .$$

Proof of Prop. 1. By Def. 3, the rehearsal graph G is composed of maximal bi-directional cliques $\{C_i\}$ that permit a topological ordering. We can thus view G as a directed acyclic graph (DAG) $\mathcal{G}_C$ where each node represents a clique C_i .

Meanwhile, according to Def. 4, the actionable set $\mathbf{A}$ is a union of disjoint cliques, and $\mathbf{U}$ consists of all variables within cliques that are ancestors of $\mathbf{A}$ in $\mathcal{G}_C$ belonging to $\mathbf{Z}$. Recall from Sec. 1 that $\mathbf{X}$ is prior than $\mathbf{Z}$ in rehearsal learning, this topological structure ensures that $\mathbf{U} \cup \mathbf{X}$ satisfies the *backdoor criterion* (Pearl, 2009) relative to the pair $(\mathbf{A}, \mathbf{Y})$ that:

1. **Non-descendance:** By definition, $\mathbf{U} \cup \mathbf{X}$ contains only ancestors of $\mathbf{A}$, ensuring no variable in $\mathbf{U}$ is a descendant of $\mathbf{A}$.
2. **Blocking Backdoor Paths:** Since $\mathbf{U} \cup \mathbf{X}$ includes all ancestors of $\mathbf{A}$, it blocks every path into $\mathbf{A}$ that originates from the ‘upstream’ of the graph. Consequently, $\mathbf{U}$ blocks all backdoor paths (*i.e.*, paths with an arrow pointing into $\mathbf{A}$) connecting $\mathbf{A}$ and $\mathbf{Y}$.

Applying the backdoor adjustment formula, the distribution of $\mathbf{Y}$ under the rehearsal operation $\dot{\mathbf{a}}$ (which assigns the values $\mathbf{a}$ to the variables in $\mathbf{A}$) can be identified via the observational distribution:

$$p(\mathbf{y} | \mathbf{x}, \dot{\mathbf{a}}) = \int_{\mathbf{u}} p(\mathbf{y} | \mathbf{x}, \mathbf{u}, \mathbf{a}) p(\mathbf{u} | \mathbf{x}) d\mathbf{u} .$$

Multiplying by $w_\eta(\mathbf{y})$ and integrating over $\mathbf{y}$ yields the target expectation:

$$\begin{aligned} \mathbb{E}_{\mathbf{y} \sim P(\cdot | \mathbf{x}, \dot{\mathbf{a}})}[w_\eta(\mathbf{Y})] &= \int_{\mathbf{y}} w_\eta(\mathbf{y}) p(\mathbf{y} | \mathbf{x}, \dot{\mathbf{a}}) d\mathbf{y} \\ &= \int_{\mathbf{y}} w_\eta(\mathbf{y}) \left(\int_{\mathbf{u}} p(\mathbf{y} | \mathbf{x}, \mathbf{u}, \mathbf{a}) p(\mathbf{u} | \mathbf{x}) d\mathbf{u} \right) d\mathbf{y} \\ &= \int_{\mathbf{u}} \left(\int_{\mathbf{y}} w_\eta(\mathbf{y}) p(\mathbf{y} | \mathbf{x}, \mathbf{u}, \mathbf{a}) d\mathbf{y} \right) p(\mathbf{u} | \mathbf{x}) d\mathbf{u} \\ &= \mathbb{E}_{\mathbf{u} \sim P(\cdot | \mathbf{x})}[\mathbb{E}_{\mathbf{y} \sim P(\cdot | \mathbf{x}, \mathbf{u}, \mathbf{a})}[w_\eta(\mathbf{Y})]] . \end{aligned}$$

□

C.2. Proof of Theorem 1

Lemma 1 (Vershynin, 2018, Proposition 2.1.2). *Let $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{u^2}{2}} du$ denote the CDF of the standard normal distribution. For any $t > 0$, the following inequality holds:*

$$\Phi(-t) \leq \frac{1}{\sqrt{2\pi}t} e^{-\frac{t^2}{2}} .$$

Lemma 2. *Let $\{u_k\}_{k=1}^l$ and $\{v_k\}_{k=1}^l$ be two sequences of real numbers such that $u_k, v_k \in [0, 1]$ for all $k = 1, \dots, l$. Then, the absolute difference of their products is bounded by the sum of their absolute pointwise differences, as shown in:*

$$\left| \prod_{k=1}^l u_k - \prod_{k=1}^l v_k \right| \leq \sum_{k=1}^l |u_k - v_k| .$$

Proof of Lemma 2. The result follows from a standard telescoping sum argument:

$$\prod_{k=1}^l u_k - \prod_{k=1}^l v_k = \sum_{k=1}^l \left(\prod_{i=1}^{k-1} v_i \right) (u_k - v_k) \left(\prod_{j=k+1}^l u_j \right) \leq \sum_{k=1}^l |u_k - v_k|.$$

The second inequality holds as $|u_i|, |v_i| \leq 1$, thus the magnitude of coefficients is bounded by 1. $\square$

Theorem 1. Let $\mathcal{S}$ be the convex polytope region defined in Eq. (2), and let $w_\eta(\mathbf{y})$ be the Probit surrogate of the indicator function $\mathbb{I}(\mathbf{y} \in \mathcal{S})$ as defined in Eq. (3). Assuming the conditional density is bounded, the gap between the true AUF probability and the surrogate objective satisfies:

$$\left| \mathbb{P}(\mathbf{Y} \in \mathcal{S} \mid \mathbf{x}, \dot{\mathbf{a}}) - \mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \dot{\mathbf{a}})} [w_\eta(\mathbf{y})] \right| = \mathcal{O} \left(\frac{\sqrt{\ln \eta}}{\eta} \right).$$

Proof of Thm. 1. Let the pointwise approximation error be denoted by $\Delta(\mathbf{y}) = |\mathbb{I}(\mathbf{y} \in \mathcal{S}) - w_\eta(\mathbf{y})|$, where the indicator function can be written as the product of indicators for each linear constraint: $\mathbb{I}(\mathbf{y} \in \mathcal{S}) = \prod_{k=1}^l \mathbb{I}(b_k - \mathbf{m}_k^\top \mathbf{y} \geq 0)$. Similarly, the surrogate is the product of Probit functions $w_\eta(\mathbf{y}) = \prod_{k=1}^l \Phi(\eta(b_k - \mathbf{m}_k^\top \mathbf{y}))$ as defined in Eq. (3). We aim to bound the probability gap. By invoking the triangle inequality for integrals, we decompose the gap into: (i) the boundary tube $\partial_\epsilon \mathcal{S}$ (points within distance ϵ of boundary); and (ii) the complement region $\mathcal{S}_\epsilon^c$ (points at least ϵ away from boundary).

$$\begin{aligned} \text{Gap} &\triangleq \left| \mathbb{P}(\mathbf{Y} \in \mathcal{S} \mid \mathbf{x}, \dot{\mathbf{a}}) - \mathbb{E}_{\mathbf{y} \sim P(\cdot \mid \mathbf{x}, \dot{\mathbf{a}})} [w_\eta(\mathbf{y})] \right| \\ &= \left| \int_{\mathbb{R}^{d_y}} (\mathbb{I}(\mathbf{y} \in \mathcal{S}) - w_\eta(\mathbf{y})) p(\mathbf{y} \mid \mathbf{x}, \dot{\mathbf{a}}) d\mathbf{y} \right| \\ &\leq \int_{\mathbb{R}^{d_y}} \Delta(\mathbf{y}) p(\mathbf{y} \mid \mathbf{x}, \dot{\mathbf{a}}) d\mathbf{y} \\ &= \int_{\partial_\epsilon \mathcal{S}} \Delta(\mathbf{y}) p(\mathbf{y} \mid \mathbf{x}, \dot{\mathbf{a}}) d\mathbf{y} + \int_{\mathcal{S}_\epsilon^c} \Delta(\mathbf{y}) p(\mathbf{y} \mid \mathbf{x}, \dot{\mathbf{a}}) d\mathbf{y}. \end{aligned} \tag{11}$$

We analyze the two integral terms separately. Without loss of generality, we assume $\|\mathbf{m}_k\|_2 = 1$ for each $\mathbf{m}_k^\top \mathbf{y} \leq b_k$ in Eq. (2), as each constraint can be rescaled accordingly.

Bound for the Boundary Region ($\partial_\epsilon \mathcal{S}$). The pointwise error can be bounded by $\Delta(\mathbf{y}) \leq 1$. Assuming the probability density $p(\mathbf{y} \mid \mathbf{x}, \dot{\mathbf{a}})$ is bounded by a constant M , the probability mass of the boundary tube depends on the volume of $\partial_\epsilon \mathcal{S}$. For a convex polytope, the volume of the ϵ -neighborhood of the boundary scales linearly with ϵ for sufficiently small ϵ w.r.t. the Euclidean metric. Thus, there exists a constant K dependent on the surface area of $\mathcal{S}$ and M such that:

$$\int_{\partial_\epsilon \mathcal{S}} \Delta(\mathbf{y}) p(\mathbf{y} \mid \mathbf{x}, \dot{\mathbf{a}}) d\mathbf{y} \leq \int_{\partial_\epsilon \mathcal{S}} M d\mathbf{y} \leq K\epsilon. \tag{12}$$

Bound for the Complement Region ($\mathcal{S}_\epsilon^c$). In this region, the distance from $\mathbf{y}$ to any defining hyperplane is at least ϵ . Let $h_k(\mathbf{y}) = b_k - \mathbf{m}_k^\top \mathbf{y}$ denote the signed margin for the k -th constraint. We distinguish two cases for $\mathbf{y} \in \mathcal{S}_\epsilon^c$ as follows:

- *Deep Inside ($\mathbf{y} \in \mathcal{S}$):* In this case, $\mathbb{I}(\mathbf{y} \in \mathcal{S}) = 1$, and $h_k(\mathbf{y}) \geq \epsilon$ for all k . The exact term $\mathbb{I}(\mathbf{y} \in \mathcal{S}) = \prod_{k=1}^l \mathbb{I}(h_k(\mathbf{y}) \geq 0) = 1$, and the surrogate $w_\eta(\mathbf{y}) = \prod_{k=1}^l \Phi(\eta h_k(\mathbf{y}))$. We apply Lemma 2 setting $u_k = 1$ and $v_k = \Phi(\eta h_k(\mathbf{y}))$:

$$\Delta(\mathbf{y}) = \left| 1 - \prod_{k=1}^l \Phi(\eta h_k(\mathbf{y})) \right| \leq \sum_{k=1}^l |1 - \Phi(\eta h_k(\mathbf{y}))| = \sum_{k=1}^l \Phi(-\eta h_k(\mathbf{y})).$$

Since $h_k(\mathbf{y}) \geq \epsilon$ and $\Phi(\cdot)$ is monotonic, $\Phi(-\eta h_k(\mathbf{y})) \leq \Phi(-\eta\epsilon)$. Thus, $\Delta(\mathbf{y}) \leq l \cdot \Phi(-\eta\epsilon)$.

- *Deep Outside ($\mathbf{y} \notin \mathcal{S}$):* In this case, $\mathbb{I}(\mathbf{y} \in \mathcal{S}) = 0$. There exists at least one constraint j such that $h_j(\mathbf{y}) \leq -\epsilon$. The exact term $\mathbb{I}(\mathbf{y} \in \mathcal{S}) = \prod_{k=1}^l \mathbb{I}(h_k(\mathbf{y}) \geq 0) = 0$, and the surrogate $w_\eta(\mathbf{y}) = \prod_{k=1}^l \Phi(\eta h_k(\mathbf{y}))$. Since $0 < \Phi(\cdot) < 1$, the product is bounded by any single factor:

$$\Delta(\mathbf{y}) = \prod_{k=1}^l \Phi(\eta h_k(\mathbf{y})) \leq \Phi(\eta h_j(\mathbf{y})) \leq \Phi(-\eta\epsilon).$$

Combining both cases, the pointwise error in $\mathcal{S}_\epsilon^c$ is globally dominated by the term involving the Gaussian tail: $\Delta(\mathbf{y}) \leq l \cdot \Phi(-\eta\epsilon)$. Now, we apply Lemma 1 setting $t = \eta\epsilon > 0$:

$$\sup_{\mathbf{y} \in \mathcal{S}_\epsilon^c} \Delta(\mathbf{y}) \leq l \cdot \Phi(-\eta\epsilon) \leq l \cdot \frac{1}{\sqrt{2\pi\eta\epsilon}} e^{-\frac{(\eta\epsilon)^2}{2}}.$$

The integral over the complement region is therefore bounded by:

$$\int_{\mathcal{S}_\epsilon^c} \Delta(\mathbf{y}) p(\mathbf{y} \mid \mathbf{x}, \mathbf{a}) d\mathbf{y} \leq \frac{l}{\sqrt{2\pi\eta\epsilon}} e^{-\frac{\eta^2\epsilon^2}{2}}. \quad (13)$$

Total Bound. Combining Eq. (12) and Eq. (13), the total gap in Eq. (11) is bounded by:

$$\text{Gap} \leq K\epsilon + \frac{C}{\eta\epsilon} e^{-\frac{\eta^2\epsilon^2}{2}},$$

where $C = l/\sqrt{2\pi}$. Choosing $\epsilon = \frac{\sqrt{2\ln\eta}}{\eta}$, we have:

- The boundary term becomes $K \frac{\sqrt{2\ln\eta}}{\eta}$.
- The tail term becomes $\frac{C}{\sqrt{2\ln\eta}} e^{-\ln\eta} = \frac{C}{\sqrt{2\ln\eta}} \frac{1}{\eta}$, which decays faster than the boundary term.

Consequently, the dominant term determines the convergence rate:

$$\text{Gap} \leq \mathcal{O}\left(\frac{\sqrt{\ln\eta}}{\eta}\right).$$

□

C.3. Proof of Theorem 2

Lemma 3 (Song et al., 2009, Theorem 6). *Let $\mathcal{V}$ and $\mathcal{Q}$ be the domains of the conditioning variable v and the target variable q , respectively. Let $\mathcal{H}_{k_q}$ denote the RKHS associated with the target domain. Consider the empirical conditional mean embedding $\hat{\mu}_{q|v}$ estimated from N samples using kernel ridge regression (KRR) with regularization parameter λ . Assuming the kernel k_v on $\mathcal{V}$ is bounded and the true embedding $\mu_{q|v}$ lies in the range of the covariance operator, for a fixed $v \in \mathcal{V}$, the estimation error satisfies the following bound in probability:*

$$\|\hat{\mu}_{q|v} - \mu_{q|v}\|_{\mathcal{H}_{k_q}} = \mathcal{O}_p\left(\frac{1}{\sqrt{N\lambda}} + \sqrt{\lambda}\right).$$

Lemma 4 (Grünwälder et al., 2012b, Lemma 2.1). *Under standard regularity assumptions on the marginal distribution and kernels defined on $\mathcal{V}$ and $\mathcal{Q}$, the empirical conditional mean embedding converges uniformly over the conditioning domain $\mathcal{V}$. Specifically, with regularization parameter λ , the following uniform bound holds in probability:*

$$\sup_{v \in \mathcal{V}} \|\hat{\mu}_{q|v} - \mu_{q|v}\|_{\mathcal{H}_{k_q}} = \mathcal{O}_p\left(\sqrt{\lambda} + \frac{1}{\sqrt{N\lambda^3}}\right).$$

Based on these lemmas, we establish the consistency and finite-sample error bound for our nested estimator.

Theorem 2. *Let N be the sample size of the observational training data, and let $\lambda_h, \lambda_x > 0$ denote the regularization parameters for the inner (Stage 1) and outer (Stage 2) KRR estimators. Suppose that w_η is bounded in the RKHS norm, i.e., $\|w_\eta\|_{\mathcal{H}_{k_y}} \leq C_w$. For any context $\mathbf{x}$ and alteration $\mathbf{a}$, the estimation error $\Delta_{est} \triangleq \left| \hat{J}_\eta(\mathbf{a}; \mathbf{x}) - J_\eta(\mathbf{a}; \mathbf{x}) \right|$ satisfies:*

$$\Delta_{est} = \mathcal{O}_p\left(\sqrt{\lambda_x} + \sqrt{\lambda_h} + \frac{1}{\sqrt{N\lambda_x}} + \frac{1}{\sqrt{N\lambda_h^3}}\right).$$

Proof of Thm. 2. Let $\psi(\mathbf{h}) \triangleq \mathbb{E}_{\mathbf{y} \sim P(\cdot|\mathbf{h})}[w_\eta(\mathbf{Y})]$ denote the true inner expectation function, and $\hat{\psi}(\mathbf{h}) \triangleq \hat{\mathbb{E}}_{\mathbf{y} \sim P(\cdot|\mathbf{h})}[w_\eta(\mathbf{Y})]$ denote its empirical estimator obtained from the first-stage KRR. Note that for fixed context $\mathbf{x}$ and alteration $\mathbf{a}$, both can be viewed as functions of $\mathbf{u}$. We aim to bound the absolute error denoted by Δ :

$$\Delta = \left| \hat{\mathbb{E}}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}] - \mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\psi] \right|.$$

To analyze the convergence, we introduce an intermediate term $\mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}(\mathbf{x}, \mathbf{u}, \mathbf{a})]$, representing the expected value of the estimated inner function under the true outer distribution. By the triangle inequality, we decompose the total error into:

$$\begin{aligned} \Delta &= \left| \hat{\mathbb{E}}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}] - \mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}] + \mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}] - \mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\psi] \right| \\ &\leq \underbrace{\left| \hat{\mathbb{E}}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}] - \mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}] \right|}_{\text{Term I: Outer Estimation Error}} + \underbrace{\left| \mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}] - \mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\psi] \right|}_{\text{Term II: Inner Estimation Error}}. \end{aligned}$$

Analysis of Term I (Outer Estimation Error): Term I captures the error arising from estimating the expectation of the fixed function $\hat{\psi}$ (conditioned on Stage 1 data) using the finite sample estimate of the outer distribution. Recall that our estimator is derived as $\hat{\mathbb{E}}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}] = \langle \hat{\psi}, \hat{\mu}_{\mathbf{U}|\mathbf{x}} \rangle_{\mathcal{H}_{k_u}}$, where $\hat{\mu}_{\mathbf{U}|\mathbf{x}}$ is the empirical CME of $\mathbf{u}$ given $\mathbf{x}$. Similarly, the intermediate term can be written as $\mathbb{E}_{\mathbf{u} \sim P(\cdot|\mathbf{x})}[\hat{\psi}] = \langle \hat{\psi}, \mu_{\mathbf{U}|\mathbf{x}} \rangle_{\mathcal{H}_{k_u}}$. Applying the Cauchy-Schwarz inequality:

$$\begin{aligned} \text{Term I} &= \left| \langle \hat{\psi}, \hat{\mu}_{\mathbf{U}|\mathbf{x}} - \mu_{\mathbf{U}|\mathbf{x}} \rangle_{\mathcal{H}_{k_u}} \right| \\ &\leq \|\hat{\psi}\|_{\mathcal{H}_{k_u}} \|\hat{\mu}_{\mathbf{U}|\mathbf{x}} - \mu_{\mathbf{U}|\mathbf{x}}\|_{\mathcal{H}_{k_u}}. \end{aligned}$$

The estimator $\hat{\psi}$ is obtained via KRR in the joint RKHS $\mathcal{H}_{\mathcal{H}}$. Under the decomposable kernel assumption, for any fixed $\mathbf{x}$ and $\mathbf{a}$, the section $\hat{\psi}(\mathbf{x}, \cdot, \mathbf{a})$ explicitly resides in $\mathcal{H}_{k_u}$. Moreover, its norm $\|\hat{\psi}\|_{\mathcal{H}_{k_u}}$ is bounded, which is a direct consequence of the boundedness of the regularized estimator in the joint space $\mathcal{H}_{\mathcal{H}}$. For the CME estimation error $\|\hat{\mu}_{\mathbf{U}|\mathbf{x}} - \mu_{\mathbf{U}|\mathbf{x}}\|_{\mathcal{H}_{k_u}}$, we invoke Lemma 3. With the outer regularization parameter λ_x , we have:

$$\text{Term I} = \mathcal{O}_p \left(\frac{1}{\sqrt{N\lambda_x}} + \sqrt{\lambda_x} \right).$$

Analysis of Term II (Inner Estimation Error): Term II reflects the propagated error from the inner expectation estimation.

$$\begin{aligned} \text{Term II} &= \left| \int_{\mathcal{U}} (\hat{\psi}(\mathbf{h}) - \psi(\mathbf{h})) p(\mathbf{u} | \mathbf{x}) d\mathbf{u} \right| \\ &\leq \sup_{\mathbf{u} \in \mathcal{U}} \left| \hat{\psi}(\mathbf{x}, \mathbf{u}, \mathbf{a}) - \psi(\mathbf{x}, \mathbf{u}, \mathbf{a}) \right|. \end{aligned}$$

According to reproducing property, $\psi(\mathbf{h}) = \langle w_\eta, \mu_{\mathbf{Y}|\mathbf{h}} \rangle_{\mathcal{H}_{k_y}}$ and $\hat{\psi}(\mathbf{h}) = \langle w_\eta, \hat{\mu}_{\mathbf{Y}|\mathbf{h}} \rangle_{\mathcal{H}_{k_y}}$. By Cauchy-Schwarz inequality:

$$\left| \hat{\psi}(\mathbf{h}) - \psi(\mathbf{h}) \right| \leq \|w_\eta\|_{\mathcal{H}_{k_y}} \|\hat{\mu}_{\mathbf{Y}|\mathbf{h}} - \mu_{\mathbf{Y}|\mathbf{h}}\|_{\mathcal{H}_{k_y}}.$$

To bound this uniformly for all $\mathbf{h}$, we rely on the uniform consistency result from Lemma 4. With the inner regularization parameter λ_h , we obtain:

$$\sup_{\mathbf{h}} \|\hat{\mu}_{\mathbf{Y}|\mathbf{h}} - \mu_{\mathbf{Y}|\mathbf{h}}\|_{\mathcal{H}_{k_y}} = \mathcal{O}_p \left(\sqrt{\lambda_h} + \frac{1}{\sqrt{N\lambda_h^3}} \right).$$

Thus, Term II is bounded by this uniform rate. Combining the bounds for Term I and Term II, we obtain the total bound:

$$\Delta \leq \text{Term I} + \text{Term II} = \mathcal{O}_p \left(\sqrt{\lambda_x} + \sqrt{\lambda_h} + \frac{1}{\sqrt{N\lambda_x}} + \frac{1}{\sqrt{N\lambda_h^3}} \right).$$

□

D. Experimental details

In this section, we provide details for the experiments in Sec. 4. Hyperparameters like bandwidths are selected empirically based on heuristics in the kernel literature (Gretton et al., 2012), and are available in the provided supplementary code.

D.1. Linear data

Bermuda data. The Bermuda data is an environmental dataset that records a collection of marine and biogeochemical variables measured in the Bermuda region (Courtney et al., 2017). The variables involved in the generation process include:

- Light: Light levels at the ocean bottom;
- Temp: Temperature at the ocean bottom;
- Sal: Sea surface salinity;
- DIC: Dissolved inorganic carbon in seawater;
- TA: Total alkalinity of seawater;
- Ω_A : Aragonite saturation state of seawater;
- Chla: Chlorophyll-*a* concentration at the sea surface;
- Nut: The first principal component (PC1) of NH_4 , $\text{NO}_2 + \text{NO}_3$, and SiO_4 ;
- pHsw: Seawater pH;
- CO_2 : Partial pressure of CO_2 in seawater (P_{CO_2});
- NEC: Net ecosystem calcification.

The rehearsal graph over these variables is illustrated in Fig. 6. Following Aglietti et al. (2020); Qin et al. (2023), the actionable variables that can be altered by the decision-maker are DIC, TA, Ω_A , Chla, and Nut, which can be altered into values with constraint $[-1.0, 1.0]$. The desired region of is $\mathcal{S} = \{\text{NEC} \in [0.5, 2]\}$, following the specifications in Sec. 4.

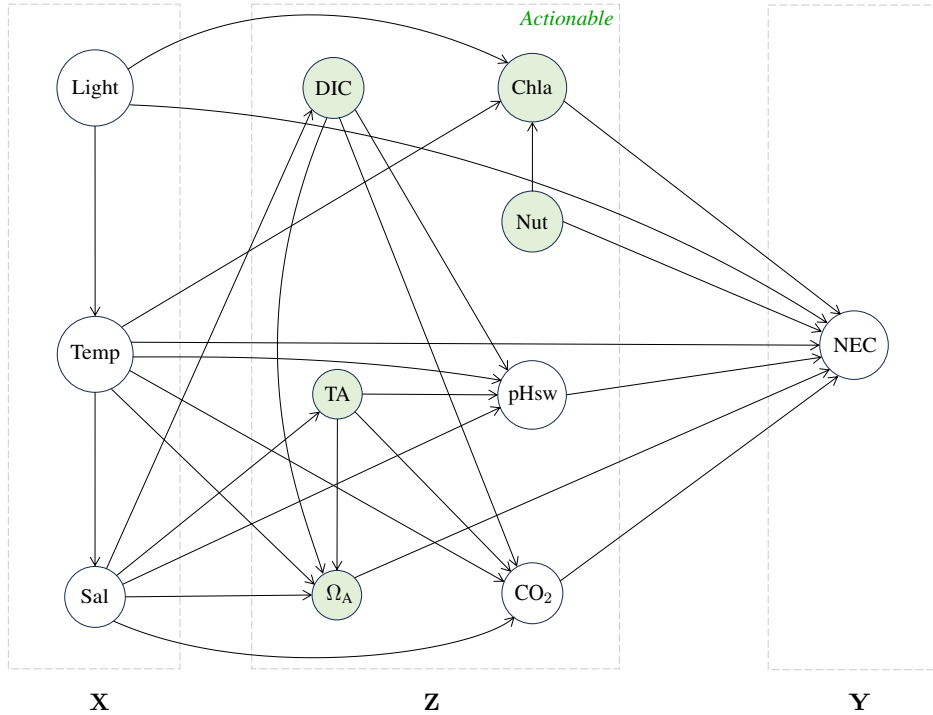


Figure 6. The graph structure of Bermuda data.

Linear synthetic data. The rehearsal graph over the variables in our linear synthetic dataset is illustrated in Fig. 7. The desired region of $\mathbf{Y}$ is specified as $\{(Y_1, Y_2) \mid 0.0 \leq Y_1 \leq 2.0, 0.0 \leq Y_2 \leq 2.0\}$, and the actionable variables are A_1 and A_2 with $A_1, A_2 \in [-3.0, 3.0]$. The generation mechanisms underlying the natural observational process are given by:

$$\left\{ \begin{array}{l} X_1 := \mathcal{N}(0, 0.1); \\ X_2 := \mathcal{N}(0, 0.1); \\ U_2 := 10.0 \times X_2 + \mathcal{N}(0, 0.1); \\ A_1 := 10.0 \times X_1 + \mathcal{N}(0, 0.1); \\ U_1 := 0.5 \times A_1 + 1.3 \times U_2 + \mathcal{N}(0, 0.1); \\ A_2 := 2.0 \times A_1 + 0.4 \times U_2 + \mathcal{N}(0, 0.1); \\ Y_1 := -1.0 \times A_1 + 0.9 \times A_2 + \mathcal{N}(0, 0.1); \\ Y_2 := 1.6 \times A_1 - 0.9 \times A_2 + \mathcal{N}(0, 0.1). \end{array} \right.$$

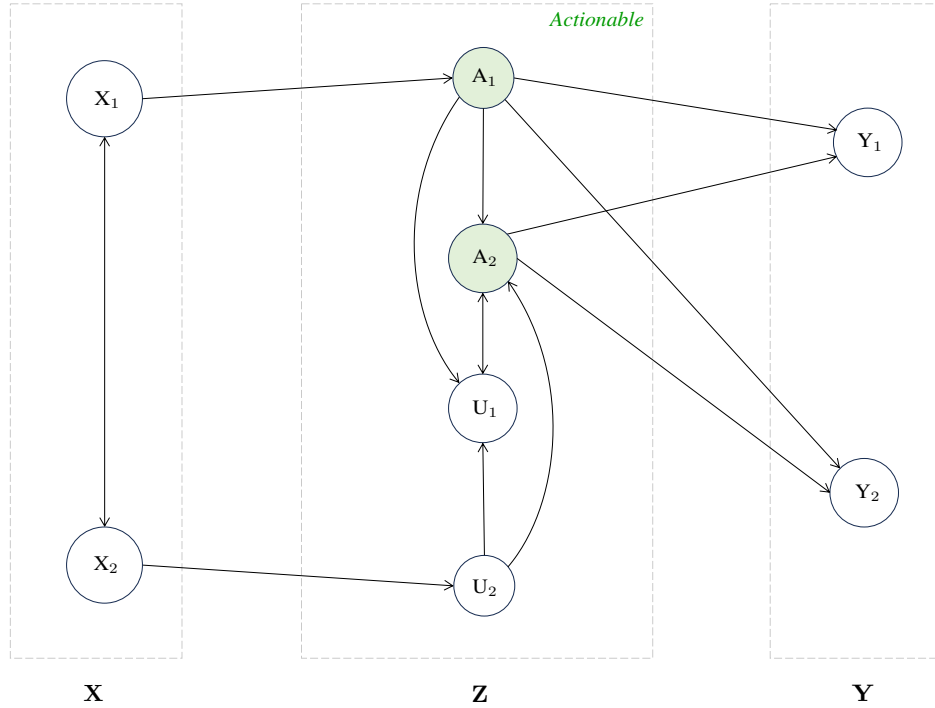


Figure 7. The graph structure of linear synthetic data.

D.2. Nonlinear data

Bank data. The rehearsal graph over the variables in our bank dataset is illustrated in Fig. 1. The actionable variables are formally defined as A_1 and A_2 (note that A_1 is excluded in our implementation to facilitate the visualization in Fig. 4). The desired region of $\mathbf{Y}$ is $\mathcal{S} = \{(Y_1, Y_2) \mid Y_1 \geq 0.6, Y_2 \geq 0.3\}$, following the specifications in Sec. 4. The variables involved in the decision process, along with the generation mechanisms underlying the natural observational process, are given by:

- X_1 : Credit Score ($X_1 := \mathcal{U}(0, 1)$);
- X_2 : Debt-to-Income Ratio ($X_2 := \mathcal{U}(0, 1)$);
- U_1 : Borrower's Industry Sector Stability Index ($U_1 \sim \text{Beta}(2, 2)$);
- A_1 : Loan Amount Granted (Excluded in implementation);
- A_2 : Interest Rate Applied ($A_2 := U_1 + 0.5X_1 + 0.5X_2 - 0.5 + \mathcal{N}(0, 0.2)$) with $A_2 \in [0.0, 1.0]$ when altering;
- Y_1 : Repayment Rate ($Y_1 := \text{Sigmoid}(2.0 \times U_1^{1.1} - 1.5 \times A_2 + 0.2 \times X_2 + 0.4) + \mathcal{N}(0, 0.05)$);
- Y_2 : Bank ROI (Return on Investment) ($Y_2 := 0.8A_2 + 0.5U_1 + \mathcal{N}(0, 0.05)$).

Nonlinear synthetic data 1. The rehearsal graph over the variables in our nonlinear synthetic dataset 1 can be described as: $X \rightarrow A_1, X \rightarrow Y; U \rightarrow A_1, U \rightarrow A_2, U \rightarrow Y; A_1 \rightarrow Y; A_2 \rightarrow Y$. The desired region of $\mathbf{Y}$ is specified as $\{Y \mid 1.5 \leq Y \leq 2.0\}$, and the actionable variables are A_1 and A_2 with $A_1, A_2 \in [-1.0, 1.0]$. The generation mechanisms underlying the natural observational process are given by:

$$\begin{cases} X := \mathcal{U}(-1, 1); \\ U := \text{Exp}(1); \\ A_1 := 0.8X + 0.2U + \mathcal{N}(0, 0.5); \\ A_2 := 0.5 \sin(U) + \mathcal{N}(0, 0.5); \\ Y := 1.5 - (A_1 - X)^2 - (A_2 - \ln(U + 1))^2 + 0.2 \sin(A_1 A_2) + \mathcal{N}(0, 0.1). \end{cases}$$

Nonlinear synthetic data 2. The rehearsal graph over the variables in our nonlinear synthetic dataset 2 is illustrated in Fig. 8. The desired region of $\mathbf{Y}$ is specified as $\{Y_1 \mid 0.9 \leq Y_1 \leq 1.5\}$, and the actionable variables are A_1 and A_2 with $A_1, A_2 \in [-1.0, 1.0]$. The generation mechanisms underlying the natural observational process are given by:

$$\begin{cases} X_1 := \mathcal{U}(0, 1); \\ X_2 := \mathcal{U}(0, 1); \\ A_1 := 1.0X_1 - 1.0X_2 + 0.2X_1^2 - 0.2X_2^2 + 0.1X_1X_2 + \mathcal{N}(0, 0.1); \\ U_1 := -1.0X_1 + 2.0X_2 + 3.0A_1 - 0.2X_1^2 + 0.4X_2^2 + 0.6A_1^2 + 0.1(X_1X_2 + X_1A_1 + X_2A_1) + \mathcal{N}(0, 0.1); \\ U_2 := -1.0X_1 + 4.0U_1 - 0.2X_1^2 + 0.8U_1^2 + 0.1X_1U_1 + \mathcal{N}(0, 0.1); \\ A_2 := -1.0X_1 - 0.5X_2 + 0.3U_1 - 0.2X_1^2 - 0.1X_2^2 + 0.06U_1^2 + 0.1(X_1X_2 + X_1U_1 + X_2U_1) + \mathcal{N}(0, 0.1); \\ Y_1 := 0.5 + \frac{1}{60} \left(0.2(X_1 + X_1^2) - 5.0(U_1 + U_1^2) - 1.0(U_2 + U_2^2) + 5.0(A_2 + A_2^2) \right) \\ + 0.5 \sum_{p_i, p_j \in \mathbf{Pa}(Y_1), i < j} p_i p_j + \mathcal{N}(0, 0.1). \end{cases}$$

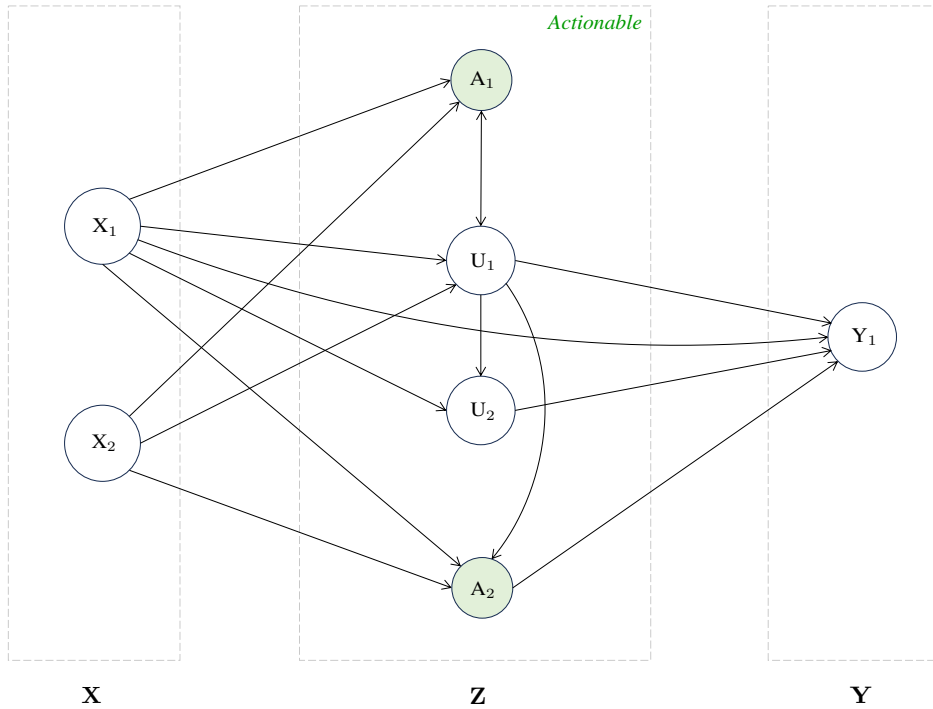


Figure 8. The graph structure of nonlinear synthetic data 2.